

Relaxations of 3×3 Magic Squares of Squares

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1 Introduction

In this paper, we will discuss our findings in studying the magic square of squares problem.

Before diving into our research, we will provide a little background on what we are referring to by a 3×3 magic square of squares, why it matters, and how our approach will differ from previous literature.

1.1 What is a magic square?

A magic square is a grid of integers whose rows, columns, main diagonal, and antidiagonal all add to the same value. Formally, we can say that $\forall n \in \mathbb{N} \ M \in \mathbb{Z}^{n \times n}$ is called “magic” if, for some $S \in \mathbb{Z}$, the following equations hold:

$$\text{Rows:} \quad \forall i \in \{1, 2, \dots, n\} \quad \sum_{j=1}^n M_{ij} = S \quad (1)$$

$$\text{Columns:} \quad \forall j \in \{1, 2, \dots, n\} \quad \sum_{i=1}^n M_{ij} = S \quad (2)$$

$$\text{Diagonals:} \quad \sum_{i=1}^n M_{ii} = S \quad (3)$$

$$\sum_{i=1}^n M_{i(n-i+1)} = S \quad (4)$$

These constraints can be trivially satisfied if all entries M_{ij} are equal or $n = 1$, and as such, our focus from this point forward will be on the more interesting case where $n > 1$ and all of the entries are distinct. Furthermore, if $n = 2$, then we get the following system of equations:

$$M_{11} + M_{12} = S \quad (1)$$

$$M_{21} + M_{22} = S \quad (2)$$

$$M_{11} + M_{21} = S \quad (3)$$

$$M_{12} + M_{22} = S \quad (4)$$

$$M_{11} + M_{22} = S \quad (5)$$

$$M_{12} + M_{21} = S \quad (6)$$

From (1) and (3), we get that $M_{12} = M_{21}$ which means that we will not have distinct entries, and, as such, our focus will be further narrowed to the case when $n \geq 3$.

There are infinitely many squares which satisfy these properties, so additional constraints were proposed in an effort to find more interesting magic squares.

1.2 What is a magic square of squares?

Now that we know what a magic square is, the next step is to impose additional constraints to make these squares more interesting. One such constraint that was proposed, possibly due to its amusing name and structure, is what is called a magic square of squares. A magic square of squares is a magic square where each entry is a distinct perfect integer square. Formally, this can be defined as follows:

$$\forall i, j \in \{1, 2, \dots, n\} \quad M_{ij} \in \{k^2 \mid k \in \mathbb{Z}\} \quad (1)$$

$$\forall i_1, j_1, i_2, j_2 \in \{1, 2, \dots, n\} \quad M_{i_1 j_1} = M_{i_2 j_2} \iff (i_1 = i_2) \wedge (j_1 = j_2) \quad (2)$$

These magic squares of squares have been the subject of research for quite some time, and just last year, it was discovered that magic squares of squares exist for all $n \geq 4$ [8].

Notably, this does not include the case when $n = 3$, which leads us finally to the topic of this paper, the Magic Square of Squares Problem.

1.3 What is the magic square of squares problem?

The magic square of squares problem is an open problem in the field of number theory that has grown in popularity since proposed with a \$100 prize by Martin Gardner in 1996 [1]. The problem concerns the question of whether there exists a magic square of squares in the case when $n = 3$. Demonstrating such a magic square of squares or proving its nonexistence would constitute a solution to this problem.

This problem has been the subject of considerable research, but, so far, no conclusive answer has been found. However, there are a few interesting discoveries that have been made:

- Many approximations have been found [1].
- Each entry of a magic square of squares must be larger than 10^{14} [1].
- Many constraints have been established on what would constitute valid entries if such a square were to exist [10].
- Research involving elliptic curves suggests that a magic square of squares is very unlikely to exist [3].

1.4 What is our contribution?

As seen above, this problem has been the subject of study for quite some time. However, reviewing the literature, it appears that the primary approaches have involved the study of elliptic curves, the generation of approximate solutions, or proving properties about the entries of such a magic square. In contrast to this, our approach will involve alternative representations and relaxations of this problem in pursuit of providing new perspectives, as well as later examining these perspectives from within the context of other algebraic structures such as finite fields rather than the ring of integers.

2 Preliminary Context

In the following section, we will discuss our research results, but, before doing so, it may be useful to establish some useful facts about the magic square of squares problem which we will list here.

2.1 3×3 Magic Square General Form

Recall that $M \in \mathbb{Z}^{n \times n}$ is called a 3×3 magic square if $n = 3$ and the row, column, and diagonal constraints defined in 1.1 are satisfied. Note, however, that these constraints are not linearly independent, and, as such, they can be simplified. Let $\mathcal{M}_n$ be the set of all $n \times n$ magic squares. One such simplification yields the following 3 variable parameterization of all 3×3 magic squares [6]:

$$\mathcal{M}_3 = \left\{ \begin{bmatrix} x+a & x-a-b & x+b \\ x-a+b & x & x+a-b \\ x-b & x+a+b & x-a \end{bmatrix} \mid x, a, b \in \mathbb{Z} \right\}$$

Note that this general form implies, as we prove in the below lemma, that 0 cannot be an entry in a 3×3 magic square of squares, so we will omit 0 from consideration from this point forward.

2.1.1 Lemma: No 3×3 Magic Square of Squares Contains 0

First, note that this general form implies that the sum S is always three times the middle entry x .

For the sake of contradiction, assume that the center entry of a 3×3 magic square of squares was 0. In this case, S must be 0. However, each entry of this magic square must be non-negative by virtue of being a perfect square, and, thus, all entries must be 0 which contradicts the distinctness requirement for entries of a magic square of squares. Thus, the center entry of a 3×3 magic square of squares cannot be 0.

For the sake of contradiction, assume that a non-center entry of a 3×3 magic square of squares was 0. There must be some row/column/diagonal sum which includes this entry 0, the center entry x , and some other entry y . This gives us the following equation $x+y = 3x$ which implies that $y = 2x$. However, recall that x is a strictly positive perfect square as we showed above, but twice a positive perfect square cannot be a perfect square, so y cannot be a perfect square. This contradicts the requirement that all entries of a magic square of squares are perfect squares. Thus, the non-center entries of a 3×3 magic square of squares cannot be 0.

Thus, since neither the center nor any non-center entry of a 3×3 magic square of squares can be 0, then no entry of a 3×3 magic square of squares can be 0.

2.2 Pythagorean Numbers

Let $u, v, w \in \mathbb{Z}$. If $u^2 + v^2 = w^2$, then (u, v, w) is called a Pythagorean triple, the corresponding triangle with side lengths u , v and hypotenuse w is called a Pythagorean triangle, and the area of this triangle $\frac{1}{2}uv$ is called a Pythagorean number.

It is known that Pythagorean numbers can be enumerated, without duplication, by using the following parameterization [7]:

For $m, n, k \in \mathbb{N}$ with $m > n$ and $\gcd(m, n) = 1$:

$$u = k \cdot (2mn)$$

$$v = k \cdot (m^2 - n^2)$$

$$w = k \cdot (m^2 + n^2)$$

Throughout our research, we have come across several properties of Pythagorean numbers, but there is one property we discovered that we had not previously seen and which we will prove below:

2.2.1 Theorem 1: Pythagorean Number Factorization

Let $n, a, b, c \in \mathbb{N}$. n can be factored as $a(c-a)$ and $b(c+b)$ if and only if n is a Pythagorean number. Note that $a(c-a) = \sum_{i=0}^{a-1} (c-1-2i)$ and $b(c+b) = \sum_{i=1}^b (c-1+2i)$, so Pythagorean numbers are the common difference of a certain 3-term arithmetic progression with middle value $c-1$.

Proof: First, suppose n is a Pythagorean number with associated triple (u, v, w) , that is $n = \frac{uv}{2}$. Let $a = \frac{1}{2}(w+u-v)$, $b = \frac{1}{2}(u+v-w)$, and $c = w$.

$$\begin{aligned} & a(c-a) = b(c+b) \\ \iff & \frac{1}{2}(w+u-v)(w-\frac{1}{2}(w+u-v)) = \frac{1}{2}(u+v-w)(w+\frac{1}{2}(u+v-w)) \text{ by construction} \\ \iff & (w+u-v)(2w-(w+u-v)) = (u+v-w)(2w+(u+v-w)) \\ \iff & (w+u-v)(w-u+v) = (u+v-w)(w+u+v) \\ \iff & w^2 - wu + wv + wu - u^2 + uv - wv + uv - v^2 = wu + u^2 + uv + wv + uv + v^2 - w^2 - wu - wv \\ \iff & w^2 - u^2 - v^2 + 2uv = u^2 + v^2 - w^2 + 2uv \\ \iff & w^2 - u^2 - v^2 = u^2 + v^2 - w^2 \\ \iff & 0 = 0 \text{ by the definition of a Pythagorean triple} \end{aligned}$$

Thus, a Pythagorean number n must have this factorization.

Next, suppose $n \in \mathbb{N}$ can be factored as $a(c-a)$ and $b(c+b)$. Let $u = a+b$, $v = b-a+c$, and $w = c$. Let's verify that this is a Pythagorean triple.

$$\begin{aligned} & u^2 + v^2 = w^2 \\ \iff & (a+b)^2 + (b-a+c)^2 = c^2 \\ \iff & a^2 + b^2 + 2ab + b^2 - ba + bc - ab + a^2 - ac + cb - ca + c^2 = c^2 \\ \iff & 2a^2 + 2b^2 + 2bc - 2ac + c^2 = c^2 \\ \iff & a^2 - ac + b^2 + bc = 0 \\ \iff & -a(c-a) + b(b+c) = 0 \\ \iff & 0 = 0 \text{ by the construction of } n \end{aligned}$$

Now, to verify that n is the Pythagorean number of this triple:

$$\begin{aligned}
& \frac{1}{2}uv \\
= & \frac{1}{2}(a+b)(b-a+c) \\
= & \frac{1}{2}(ab - a^2 + ac + b^2 - ab + cb) \\
= & \frac{1}{2}(ac - a^2 + b^2 + cb) \\
= & \frac{1}{2}(a(c-a) + b(b+c)) \\
= & \frac{1}{2}(n+n) \text{ by the construction of } n \\
= & n
\end{aligned}$$

So, n is a Pythagorean number if it has this factorization.

2.3 Congrua / Arithmetic Progression of Squares

Let $x \in \{i^2 \mid i \in \mathbb{Z}\}$ and $a \in \mathbb{N}$. If $x - a$ and $x + a$ are perfect integer squares, then a is called a congruum. That is, a congruum is the common difference between three square numbers in arithmetic progression. Note that there does not exist an arithmetic progression of four squares [2].

Looking at the above parameterization of 3×3 magic squares in 2.1, we can observe that, in the case of a magic square of squares, the center entry x must be a perfect integer square. Without loss of generality, assume that $a > b > 0$, and this gives us that a , b , $a + b$, and $a - b$ must be congrua for x by the definition of congrua stated above. Thus, finding a magic square of squares is equivalent to finding two congrua a, b for a perfect integer square x such that both $a + b$ and $a - b$ are congrua for x .

If one researches congrua, they may encounter the following parameterization:

$$\begin{aligned}
& \text{For } m, n \in \mathbb{N} \text{ with } m > n : \\
& x = (m^2 + n^2)^2 \\
& a = 4mn(m^2 - n^2)
\end{aligned}$$

If following this procedure, a will be a congruum for x [5]. However, as discovered by an anonymous user on Wikipedia, this does not generate all possible congrua such as $a = 49896$ for $x = 255^2$ because 255 is not a sum of squares.

A congruum is related to the concept of a congruent number, which is defined as the area of a right triangle with rational sides, as every congruum is congruent and every congruent number is a congruum multiplied by the square of a rational number [4]. It is known that a is a congruum if and only if it is four times the area of a Pythagorean triangle [9]. With the definition above in 2.2, any Pythagorean number has at least one associated Pythagorean triple (u, v, w) . Note that the triple $(2u, 2v, 2w)$ is also Pythagorean and thus yields a Pythagorean number that is four times greater than the original. Thus, since a congruum is four times the area of a Pythagorean triangle, we know that all congrua are Pythagorean numbers.

This fact leads us to construct a parameterization that resolves the issues of the previous one and indeed enumerates all congrua.

2.3.1 Theorem 2: Congrua Enumeration

The following parameterization enumerates all congrua:

For $m, n, k \in \mathbb{N}$ with $m > n$ and $\gcd(m, n) = 1$:

$$\begin{aligned} x &= (k \cdot (m^2 + n^2))^2 \\ a &= 4k^2 \cdot mn(m^2 - n^2) \end{aligned}$$

Proof: First, start with the aforementioned enumeration for Pythagorean triples:

For $m, n, k \in \mathbb{N}$ with $m > n$ and $\gcd(m, n) = 1$:

$$\begin{aligned} u &= k \cdot (2mn) \\ v &= k \cdot (m^2 - n^2) \\ w &= k \cdot (m^2 + n^2) \end{aligned}$$

If this enumeration covers all triples, then the following enumeration, which enumerates the Pythagorean numbers for each of these triples, must enumerate all Pythagorean numbers p with hypotenuse w .

For $m, n, k \in \mathbb{N}$ with $m > n$ and $\gcd(m, n) = 1$:

$$\begin{aligned} w &= k \cdot (m^2 + n^2) \\ p &= \frac{1}{2}k^2 \cdot (2mn)(m^2 - n^2) \end{aligned}$$

Let p be our Pythagorean number. Recall from Theorem 1 that $a = \frac{1}{2}(w+u-v)$, $b = \frac{1}{2}(u+v-w)$, and $c = w$ forms a 3-term arithmetic progression with middle value $c - 1$ and common difference written as either $a(c - a) = \sum_{i=0}^{a-1} (c - 1 - 2i)$ or $b(c + b) = \sum_{i=1}^b (c - 1 + 2i)$. Also note the identity $k^2 = \sum_{i=1}^k (2i - 1)$. When c is even, this gives us the following:

$$\begin{aligned} \left(\frac{c}{2}\right)^2 - p &= \sum_{i=1}^{\frac{c}{2}} (2i - 1) - \sum_{i=0}^{a-1} (c - 1 - 2i) = \sum_{i=1}^{\frac{c}{2}-a} (2i - 1) = \left(\frac{c}{2} - a\right)^2 \\ \left(\frac{c}{2}\right)^2 + p &= \sum_{i=1}^{\frac{c}{2}} (2i - 1) + \sum_{i=1}^b (c - 1 + 2i) = \sum_{i=1}^{\frac{c}{2}+b} (2i - 1) = \left(\frac{c}{2} + b\right)^2 \end{aligned}$$

Thus, we can change the first term in our enumeration from the hypotenuse w to the middle square x of the congruum as follows:

For $m, n, k \in \mathbb{N}$ with $m > n$ and $\gcd(m, n) = 1$:

$$\begin{aligned} x &= \left(\frac{1}{2}k \cdot (m^2 + n^2)\right)^2 \\ p &= \frac{1}{2}k^2 \cdot (2mn)(m^2 - n^2) \end{aligned}$$

Now taking the above enumeration and multiplying p by 4 (while adjusting x accordingly) allows us to enumerate all numbers that are 4 times as large as a Pythagorean number, i.e., congrua. We do this to skip non-congrua.

For $m, n, k \in \mathbb{N}$ with $m > n$ and $\gcd(m, n) = 1$:

$$\begin{aligned} x &= (k \cdot (m^2 + n^2))^2 \\ p &= 2k^2 \cdot (2mn)(m^2 - n^2) \end{aligned}$$

This is just the enumeration that we wanted to show.

2.4 Congrua Residue Graph

Here we will introduce a new concept called the Congrua Residue Graph. We create a countably infinite directed graph $G = (V, E)$. We let $V = \{i^2 \mid i \in \mathbb{N}\}$, and, for each congruum x, a , we add edge $(x, x + a)$ and $(x, x - a)$ in E .

The purpose of introducing this structure is to permit an alternative way of investigating this problem through analyzing a weaker condition than a magic square of squares. This leads us to the following result.

2.4.1 Theorem 3: CRG Antitransitivity Implies MSS Nonexistence

If the congrua residue graph is antitransitive, then no magic square of squares exists.

Proof: Suppose G were antitransitive. Then $\nexists (u, v), (v, w), (u, w) \in E$. By construction, this tells us that there are no congrua a, b for u and c for v such that $u + a = v$ or $u - a = v$, $v + c = w$ or $v - c = w$, and $u + b = w$ or $u - b = w$. This, among other things, tells us that, for any two congrua a, b for $u \in V$, $a + b$ is not a congruum for u since, otherwise, there would be edges $(u, u + a), (u + a, u + a + b), (u, u + a + b)$. Hence, the antitransitivity of the congrua residue graph necessarily entails the nonexistence of a magic square of squares and would be a much stronger condition.

3 Problem Taxonomy

We can apply these ideas to develop a taxonomy for the magic square of squares problem. Firstly, a magic square of squares exists if and only if there exist two congrua $a > b$ for x where $a + b$ and $a - b$ are also congrua for x . From this, we construct the following necessary conditions for the existence of a magic square of squares.

1. From the parametrization of 3×3 magic squares, if a magic square of squares exists, there must exist two congrua $a > b$ for some x such that $a + b$ or $a - b$ is a congruum for x .
2. Since congrua are just four times the area of a Pythagorean number, if a magic square of squares exists, there must exist three Pythagorean triples $(u_1, v_1, w), (u_2, v_2, w), (u_3, v_3, w)$ such that $u_1v_1 + u_2v_2 = u_3v_3$ or $u_1v_1 - u_2v_2 = u_3v_3$.
3. Looking at the congrua residue graph $G = (V, E)$, if a magic square of squares exists, the graph must not be antitransitive, i.e., $\exists u, v, w \in V$ such that $\{(u, v), (v, w), (u, w)\} \subset E$.

4 Finite Fields

The above constructions are defined over the ring of integers $\mathbb{Z}$ or the semiring $\mathbb{N}$. However, we could instead consider looking at solutions to this problem within other algebraic structures such as finite fields with the goal of extrapolating the results back to $\mathbb{Z}$ or $\mathbb{N}$ respectively. See the Appendix (CRG) for an example of congrua residue graphs in finite fields.

For the purposes of this paper, we will define the following sets:

- Magic Squares:
 - Let $\mathcal{S}_p$ be the set of 3×3 magic squares whose entries are (possibly non-distinct) residues in $\mathbb{F}_p$.
 - Let $\mathcal{S}_p^! \subseteq \mathcal{S}_p$ be the subset of squares where each entry is distinct.
 - Let $\mathcal{S}_p^{!*} \subseteq \mathcal{S}_p^!$ be the subset of squares where each entry is nonzero.
- Congrua:
 - Let $\mathcal{C}_p$ be the set of elements $(a, x) \in \mathbb{F}_p^2$ such that $a \neq 0$ and $x, x + a$, and $x - a$ are residues in $\mathbb{F}_p$.
 - Let $\mathcal{C}_p^* \subseteq \mathcal{C}_p$ contain the elements $(a, x) \in \mathcal{C}_p$ where $x \neq 0$ and $x + a \neq 0$.
 - Let $\mathcal{AC}_p$ be the set of elements $(a, b, x) \in \mathbb{F}_p^3$ such that $(a, x) \in \mathcal{C}_p$, $(b, x) \in \mathcal{C}_p$, and $(a + b, x) \in \mathcal{C}_p$.
 - Let $\mathcal{AC}_p^! \subseteq \mathcal{AC}_p$ be the set of elements $(a, b, x) \in \mathcal{AC}_p$ such that $a \neq b$.
 - Let $\mathcal{AC}_p^{!*} \subseteq \mathcal{AC}_p^!$ be the set of elements $(a, b, x) \in \mathcal{AC}_p^!$ such that $(a, x) \in \mathcal{C}_p^*$, $(b, x) \in \mathcal{C}_p^*$, and $(a + b, x) \in \mathcal{C}_p^*$.
- Transitivity:
 - Let $\mathcal{T}_p$ be the set of elements $(a, b, x, y) \in \mathbb{F}_p^4$ such that $(a, x) \in \mathcal{C}_p$, $(b, y) \in \mathcal{C}_p$, and $(a + b, x) \in \mathcal{C}_p$.
 - Let $\mathcal{T}_p^* \subseteq \mathcal{T}_p$ be the set of elements $(a, b, x, y) \in \mathcal{T}_p$ such that $(a, x) \in \mathcal{C}_p^*$, $(b, y) \in \mathcal{C}_p^*$, and $(a + b, x) \in \mathcal{C}_p^*$.
 - Let $\mathcal{T}_p^{*!} \subseteq \mathcal{T}_p^*$ be the set of elements $(a, b, x, y) \in \mathcal{T}_p^*$ such that $a \neq b$.

We found that magic squares of squares indeed exist within finite fields as small as $\mathbb{F}_{59}$ if 0 is not permitted as an entry and $\mathbb{F}_{29}$ if 0 is permitted. From here, it seemed reasonable to attempt to count the number of such squares that exist within these fields (e.g. 56 0-inclusive squares in $\mathbb{F}_{29}$ and 232 both 0-inclusive and 0-exclusive squares in $\mathbb{F}_{59}$). Empirically, it was observed that the number of magic squares of squares within $\mathbb{F}_p$ was divisible by $p - 1$. However, it was hard to discern many other patterns within the data.

As a result, we decided to look at congrua and congrua residue graphs in the hopes that this might help us to discover whatever pattern there may be. We set about calculating the cardinality of the above Congrua and Transitivity sets. Coming up with a counting function for many of these seemed out of reach. However, after calculating the cardinality of the simplest sets $\mathcal{C}_p$ and $\mathcal{C}_p^*$, we noticed that they followed a formula:

$$|\mathcal{C}_p^*| = \frac{1}{2}(p-1) \lfloor \frac{1}{4}(p-3) \rfloor = \begin{cases} \frac{1}{2}(p-1)\frac{1}{4}(p-5) & p \equiv 1 \pmod{4} \\ \frac{1}{2}(p-1)\frac{1}{4}(p-3) & p \equiv 3 \pmod{4} \end{cases}$$

$$|\mathcal{C}_p| = \begin{cases} \frac{1}{2}(p-1)\frac{1}{4}(p+3) & p \equiv 1 \pmod{8} \\ \frac{1}{2}(p-1)\frac{1}{4}(p-3) & p \equiv 3 \pmod{8} \\ \frac{1}{2}(p-1)\frac{1}{4}(p-1) & p \equiv 5 \pmod{8} \\ \frac{1}{2}(p-1)\frac{1}{4}(p+1) & p \equiv 7 \pmod{8} \end{cases}$$

We also noticed that nodes within the congrua residue graph had identical degree, and thus, we can calculate congrua for each residue using these formulas by removing the $\frac{1}{2}(p-1)$ term. See the Appendix (Cardinality Tables) to see the exact values that we calculated.

These results have yet to be proven, but, if correct, they illustrate a structure to congrua, at least within finite fields, and perhaps bring us one step closer to being able to count magic squares of squares within finite fields.

5 Conclusion

In this paper, we have defined the magic square of squares problem and tackled it from several different angles. Firstly, we constructed necessary conditions for a magic square of squares in search of simpler problems with solutions that could extrapolate to the magic square of squares. Secondly, we focused in on the algebraic structure of finite fields and collected empirical data in an effort to analyze these problems within a simpler environment. Finally, we aimed to construct formulas from these results in order to describe the behavior that we saw.

Future work could follow a few different approaches:

- Aim to find additional necessary or sufficient conditions for a magic square of squares to exist in general.
- Try finding and analyzing the behavior of magic squares of squares within algebraic structures other than the finite fields.
- Attempt to further analyze the behavior of these squares within finite fields with the goal of discovering formulas that can better describe this behavior.

While it is still unknown whether or not a 3×3 magic square of squares exists, we hope that our results will assist in answering this question.

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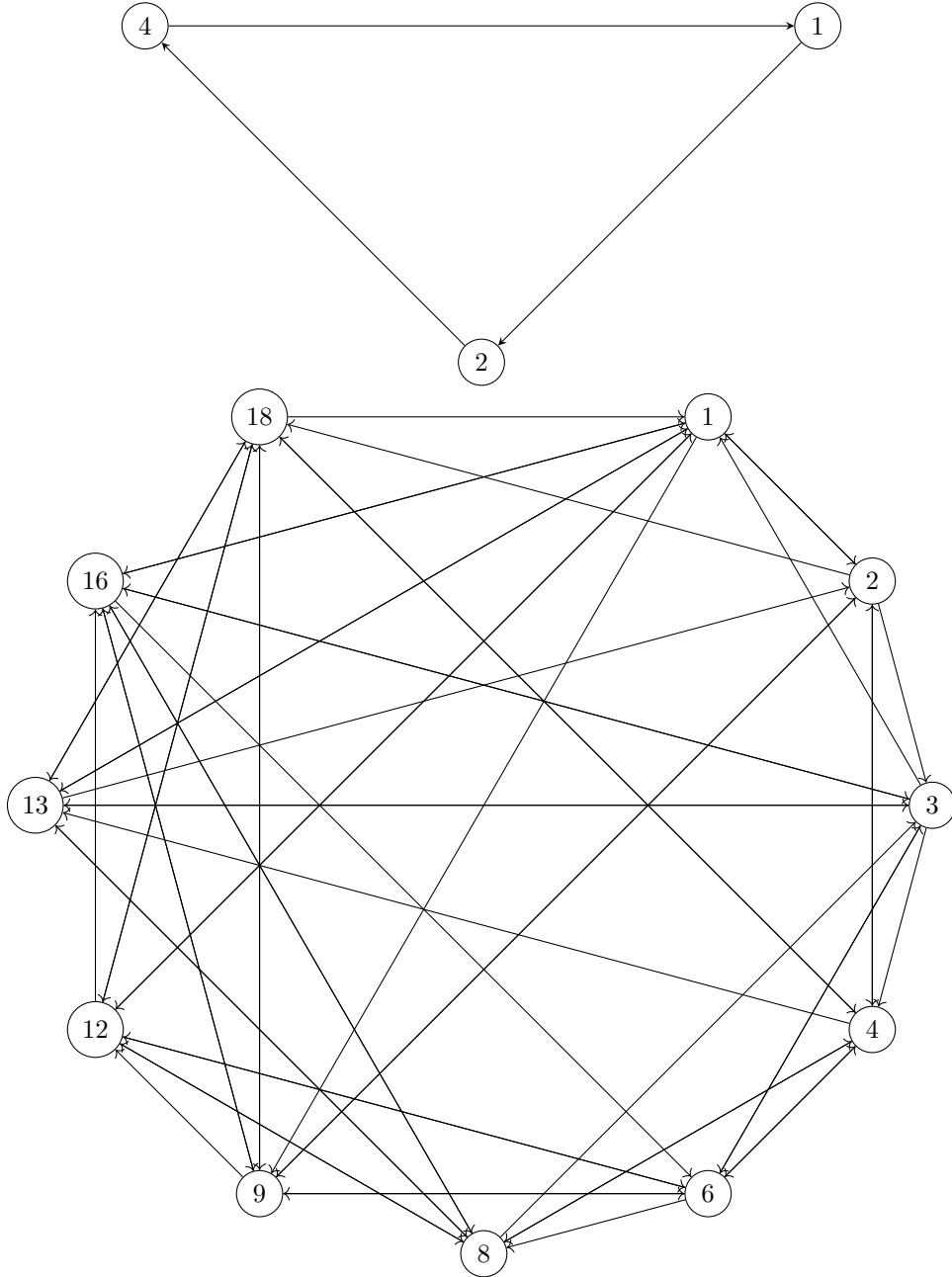
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6 Appendix

6.1 CRG

Below is an example of the congrua residue graphs for the finite fields $\mathbb{F}_7$ and $\mathbb{F}_{23}$ respectively:



6.2 Cardinality Tables

Below is a table which shows a sampling of the set cardinalities that we calculated.

p	$ \mathcal{S}_p $	$ \mathcal{S}_p^I $	$ \mathcal{S}_p^{I*} $	$ \mathcal{AC}_p $	$ \mathcal{AC}_p^I $	$ \mathcal{AC}_p^{I*} $	$ \mathcal{T}_p $	$ \mathcal{T}_p^* $	$ \mathcal{T}_p^{I*} $	$ \mathcal{C}_p $	$ \mathcal{C}_p^* $
2	8	0	0	0	0	0	0	0	0	2	0
3	2	0	0	0	0	0	0	0	0	0	0
5	7	0	0	0	0	0	0	0	0	2	0
7	16	0	0	0	0	0	3	0	0	6	3
11	26	0	0	0	0	0	5	5	5	10	10
13	43	0	0	12	12	0	12	0	0	18	12
17	105	0	0	24	16	0	88	16	16	40	24
19	82	0	0	0	0	0	36	36	36	36	36
23	188	0	0	66	44	33	132	77	66	66	55
29	267	56	0	84	84	0	252	168	168	98	84
31	256	0	0	0	0	0	420	315	315	120	105
37	415	72	0	360	360	216	540	396	396	162	144
41	621	40	0	300	240	120	1120	580	540	220	180
43	526	0	0	126	84	126	882	882	840	210	210
47	668	0	0	414	368	345	1403	1150	1127	276	253
53	1015	104	0	624	520	312	1846	1534	1430	338	312
59	1190	232	232	1218	1160	1218	2639	2639	2581	406	406
61	1411	240	0	1500	1380	1080	2940	2520	2400	450	420
67	1354	0	0	1188	1056	1188	3795	3795	3663	528	528
71	2276	280	0	2730	2520	2205	5075	4480	4305	630	595
73	2917	720	288	3204	3024	2088	6588	4752	4680	684	612
79	2536	624	0	3276	3120	2808	7137	6396	6240	780	741
83	2830	328	328	3690	3444	3690	7421	7421	7175	820	820
89	3213	704	0	3564	3344	2376	11264	8492	8316	1012	924
97	5617	1728	1152	7440	7008	5664	14496	11184	10896	1200	1104
101	4351	600	0	6600	6200	5400	14250	13050	12650	1250	1200
103	4948	1224	0	7650	7344	6732	16371	15096	14790	1326	1275
107	4294	848	848	7950	7632	7950	16483	16483	16165	1378	1378
109	6643	1944	1296	10476	10044	9072	18198	16794	16362	1458	1404
113	7113	3472	896	9912	9744	7392	23408	18872	18760	1624	1512
127	8884	3024	504	15498	14868	13608	30240	28287	27657	2016	1953
131	7866	1560	1560	13260	12740	13260	31265	31265	30745	2080	2080
137	11901	6120	1632	18972	18496	15096	41480	34748	34340	2380	2244